

When Verification Axes Can Support Scientific Claims: Identifiability, Attainability, and Non-Compensatory Promotion

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Abstract

Verification layers for learning-based research agents are often evaluated by benchmark scores, but a score can support a competence claim only when the target is identifiable, compared interfaces can attain its terminal, and the panel is discriminating. We formalize these requirements through fibrewise Bayes risk, a donor-factorization theorem, a terminal-adaptor criterion, and data-processing and total-variation bounds. In a protected 420-case battery, a non-compensatory promotion relation made 0/360 false promotions versus 180/360 for the strongest frozen mechanism proxy, while both promoted 60/60 clean positives. The registered abstention contrast on that battery was saturated and remains a negative measurement result. On a distinct shortcut-resistant exact-axis battery, the governed pipeline selected the undetermined terminal on 30/30 eligible cases versus 0/30 for the same strongest proxy and 15/30 for the one escalation-capable mechanism; the paired difference against the strongest proxy was 1.0 (95% CI [1.0, 1.0]), while the margin against the 15/30 mechanism was 0.5. This measures terminal expressiveness under the frozen gate lattice, not general scientific judgement. On 400 exact contracts, the target-bound relation scored 400/400 versus 250/400 for a donor-complete generic product and 50/400 for a compensatory product, while an information-equivalent typed product tied at 400/400. The paired improvement over the donor-complete generic product was 0.375 with a domain-stratified 95% bootstrap interval of [0.3275, 0.4225]. These finite results support an identifiability-and-attainability boundary for scientific promotion; they do not establish naturalistic or deployed-system superiority.

Keywords: scientific verification; benchmark identifiability; attainable risk; provenance; authorization; protected evaluation; abstention.

1 Introduction

Language-model research systems increasingly retrieve sources, synthesize claims, attach citations, and run automated verification (Rasheed et al., 2026; Alvarez et al., 2026). This creates a governance problem: a claim may be correct but attributed to the wrong source; a source may support a nearby claim but not the assigned one; a checker may be non-empty yet non-discriminating or share lineage with the answer; and an evaluator may be contaminated or tampered with after the fact. In each case, a scalar confidence score can remain high even though a prerequisite for scientific authority is absent.

We study a stricter object: a *scientific-authority transition*. A proposal can be plausible and well supported but is promoted only when a registered set of content, provenance, checker, and evaluator prerequisites all pass. Unresolved prerequisites do not contribute a low score; they prevent promotion and leave the decision undetermined. This design is motivated by source-aware verification, multi-source attribution, claim-level auditability, evidence influence, iterative fact checking, evaluator-tampering benchmarks, and search-time

A large language model was used to assist with manuscript editing and preparation of the review package.

contamination work (Alvarez et al., 2026; Li et al., 2024; Rasheed et al., 2026; Faizan & Alharthi, 2026; Xie et al., 2024; Atinafu & Cohen, 2026; Wang et al., 2026c; Sadeghi et al., 2026). Recent academic-integrity and agentic-abstention benchmarks also make explicit that declining to complete an unsupported action can itself be the correct behavior rather than a failure to optimize task success (Yang et al., 2026; Liu et al., 2026).

The contribution is deliberately narrower than those component techniques. We ask whether composing these requirements as a *non-compensatory, protected authority transition* materially reduces false scientific-authority promotion while preserving legitimate clean coverage. Throughout, the pipeline built to that rule is called the *governed pipeline*, so that each claim reads as a claim about the rule rather than about a product; Section 5 names the implementation that instantiates it.

The publication-authorizing campaign was executed only after the subject, comparator and ablation mechanisms, evaluator, statistical rules, and split-generation code were frozen. A separate 39-case live-model arm remains exploratory because post-run audit found label/denominator inconsistencies and a hidden-family execution leak; it is retained as adverse evidence and is not used to authorize, tune, or rescue the protected result.

Our main findings are: (i) the governed pipeline made 0/360 false promotions while the strongest frozen mechanism proxy made 180/360, with the paired H1 interval entirely beyond the predeclared five-point margin; (ii) both promoted 60/60 clean positives, satisfying the H2 non-inferiority guard; (iii) H3 was not supported on that battery, because every system left all eligible cases correctly undetermined on a family that no system could get wrong, while on a repaired battery whose construction clears a fourteen-probe identifiability register the same hypothesis is supported at 1.0 (95% CI [1.0, 1.0]) against the strongest comparator’s 0/30 and at 0.5 against the 15/30 of the one comparator mechanism that can escalate, which measures the ability to express an undetermined terminal rather than a finer-grained judgement; and (iv) every registered ablation increased false promotion without reducing clean coverage, with soft confidence producing the largest degradation.

2 Related work

Source ownership and attribution. Attribution evaluation across multiple candidate sources is an established measurement problem (Li et al., 2024); source-aware factuality work targets cross-source conflation directly (Alvarez et al., 2026); and scientific claim–evidence benchmarks test whether retrieved evidence supports or contradicts a claim (Javaji et al., 2025). Claim-source retrieval sharpens the same point from the retrieval side, since recovering the source a scientific statement actually came from is a different task from finding a source that would support it (Schrieder et al., 2026). Together these establish claim correctness, source ownership, and semantic support as separate coordinates, which is the premise this paper builds on.

Provenance-sensitive authorization. The distinction between relevant evidence and evidence authorized to determine an action or conclusion is already established. A controlled audit holds the task and proposition fixed while varying source authority and measures whether agent actions change (Liao, 2026). Partial Evidence Bench studies a complementary failure mode in which access control is correct but a system overstates completeness when material evidence lies outside its authorization boundary (Tallam, 2026). AuthGraph aligns execution provenance against a separately derived authorization graph at tool and parameter-source level (Wang et al., 2026b); FAVA lowers structured task constraints into evidence-backed permission graphs checked before effectful execution (Zhang et al., 2026); and cryptographically verifiable authorization formalizes bindings among principal, request, context, policy, and execution evidence (Llambí-Morillas & Fernández-Fernández, 2026). These works absorb any generic claim that provenance-aware authorization, permission graphs, or evidence-backed authorization are new here. The residual question is narrower: when the target is *scientific claim promotion*, what representation and output interface make that target attainable, and does a target-bound non-compensatory relation differ from a donor-complete generic authorization relation on prospectively frozen exact contracts?

Auditability, citation fidelity, and influence. Claim-level auditability work argues that research-agent outputs require persistent semantic provenance rather than only fluent text (Rasheed et al., 2026). ProvenAI separates citation fidelity from whether cited evidence influenced the answer path (Faizan & Alharthi, 2026). We adopt that separation: citation presence cannot substitute for evidence-use provenance.

Research-integrity provenance and declared behavior. INSPECT-AI and RIPE-KG represent the provenance of research-integrity assessments over clinical-trial publications, showing a current scientific setting in which the assessment process itself is an auditable object (Markovic et al., 2026). Behavioral Integrity Verification formalizes a complementary declared-versus-actual integrity problem for agent skills (Wu et al., 2026), and certified speculative execution shows the same instinct applied to untrusted agent actions: let the work proceed, but hold its effects behind a check that must clear before they become authoritative (Zhou et al., 2026). These are parent-domain precedents for explicit provenance and contract checking.

Iterative verification and evidence escalation. FIRE iterates retrieval and verification rather than treating a single retrieval pass as final (Xie et al., 2024). DeepSciVerify motivates escalation from an abstract-level judgment to fuller evidence when initial evidence is insufficient (Sadeghi et al., 2026). Stage-wise fact-checking benchmarks make the complementary measurement argument, that an end-to-end verdict hides which stage actually failed (Lin et al., 2026). The first two mechanisms inform frozen comparator proxies here, and the third informs the decision to report per-family and per-coordinate outcomes rather than one aggregate.

Evaluator integrity, benchmark auditing, and contamination. RewardHackingAgents treats evaluator manipulation and held-out leakage as an explicit benchmark dimension (Atinafu & Cohen, 2026). Search-time contamination work shows that browsing systems can retrieve public benchmark answers during inference (Wang et al., 2026c). BenchGuard and Automated Benchmark Auditing show that benchmark specifications, environments, ground truths, and grading logic can themselves be defective (Tu et al., 2026; Wang et al., 2026a). The Holistic Agent Leaderboard further demonstrates the value of standardized harnessing and trace inspection, including direct observation of agents searching for benchmark material rather than solving the intended task (Kapoor et al., 2025). These results motivate the protected evaluator identity, holdout custody, access telemetry, and independent audit used here.

Abstention and scientific integrity. SciIntegrity-Bench constructs academic-integrity dilemmas in which honest acknowledgement of inability is the only correct outcome, while AgentAbstain evaluates paired should-act/should-abstain tasks in executable environments (Yang et al., 2026; Liu et al., 2026). They establish abstention as an evaluation target distinct from generic task success. The object measured here is narrower: an undetermined or blocked terminal is authority-specific, produced by an unresolved or failed evidence, checker, or evaluator prerequisite rather than by a general judgement that the task should not be attempted.

3 Verification-axis identifiability and attainability

A benchmark does not measure a scientific competence merely because it assigns a label and reports an accuracy. The label must be recoverable from the intended scientific information, unavailable from construction nuisances, expressible by the systems being compared, and capable of varying over the comparison panel. These are different requirements. We formalize them before interpreting the protected results.

Let $Y_a \in \mathcal{T}$ be the correct terminal on verification axis a for a random benchmark unit, let D be the representation available to a decision rule, and let finite nonempty $\mathcal{A} \subseteq \mathcal{T}$ be the terminals that the rule is allowed to emit. For a finite benchmark distribution, the smallest attainable exact-terminal error is

$$R_a^*(D, \mathcal{A}) = 1 - \mathbb{E}_D \left[\max_{t \in \mathcal{A}} \Pr(Y_a = t \mid D) \right]. \quad (1)$$

This quantity separates a poor implementation from a task that its information or interface makes impossible.

Proposition 1 (exact axis attainability). There exists a decision rule $g(D) \in \mathcal{A}$ with zero exact-terminal error if and only if every positive-probability fibre of D contains one target terminal and that terminal belongs to $\mathcal{A}$. Equivalently, $Y_a = g(D)$ almost surely for some g with codomain $\mathcal{A}$. This is an almost-sure benchmark statement: it says nothing pointwise about worlds on null fibres.

Proof. On a fibre $D = d$, any rule must emit one terminal. It is correct on the whole fibre exactly when the conditional support of Y_a is the singleton containing that terminal. Summing the best fibrewise choices gives Equation 1, and zero risk gives the stated condition. $\square$

Corollary 1 (donor-product factorization at two authorities). Let D collect the states exposed by a product of provenance, verification, custody, freshness, and generic-authorization donors, and let Y_{promote} be the target scientific-promotion terminal. Pointwise extensional equivalence on a declared world class holds if and only if the target terminal is constant on every donor-state fibre in that class and belongs to the output alphabet. Zero error under a benchmark distribution requires only the corresponding condition on positive-probability fibres; null fibres cannot be certified by a distributional score. If a positive-probability fibre mixes target terminals, every donor-only rule has error at least

$$\sum_d \Pr(D = d) \left[1 - \max_{t \in \mathcal{A}} \Pr(Y_{\text{promote}} = t \mid D = d) \right]. \quad (2)$$

The pointwise statement is the set-theoretic factorization argument; the distributional statement is Proposition 1, and the displayed lower bound is attained by a fibrewise Bayes rule on the finite benchmark. Thus a separate module has no inherent advantage once a donor product receives a target-sufficient representation and the same decision relation; conversely, no optimization of a representation that merges differently licensed states can recover the missing distinction. This is a factorization boundary, not a claim that scientific authority must be centralized.

Corollary 2 (terminal-preserving comparator adapters). Let $V \in \mathcal{V}$ be the complete candidate-visible record supplied to an external comparator and let $X \in \mathcal{X}$ be its native output, where $\mathcal{V}$ and $\mathcal{X}$ are standard-Borel spaces. Allow X to be generated by a measurable randomized kernel from V , so that the comparator output adds no world information beyond V . A benchmark adapter is a prospectively declared measurable map $h : \mathcal{X} \rightarrow \mathcal{A}$; it is not a post-hoc relabelling rule. Its exact-terminal risk is bounded below by

$$R_a^*(X, \mathcal{A}) = 1 - \mathbb{E}_X \left[\max_{t \in \mathcal{A}} \Pr(Y_a = t \mid X) \right] \geq R_a^*(V, \mathcal{A}).$$

The first bound is attained by a measurable fixed-order Bayes adapter, and the second inequality is the data-processing consequence of $Y_a - V - X$. Thus a native output is admissible for a three-way exact-axis endpoint with zero benchmark error if and only if its positive-probability output fibres are target-pure and their targets belong to $\mathcal{A}$. For a deterministic native-output map, pointwise admissibility on a declared world class requires the same purity on every native-output fibre, including benchmark-null fibres. In particular, a binary native alphabet cannot attain an endpoint on which all three target terminals have positive probability, and a native BLOCK, “not attributable,” parse failure, or free-text insufficiency response cannot be renamed CANNOTCHECK unless the predeclared semantics-preserving map satisfies this fibre condition. When it does not, the mixed-fibre risk is an interface-attainability result and must not be reported as comparator inferiority.

Proposition 2 (claim-level indistinguishability). Let W_0 and W_1 be two scientific worlds in which a proposed competence claim has opposite truth values, and let O be the complete observable record available to an adjudicator. Write $P_i = \mathcal{L}(O \mid W_i)$ for probability measures on one common measurable record space $(\mathcal{O}, \mathcal{F}_O)$, and use the convention $\text{TV}(P_0, P_1) = \sup_{B \in \mathcal{F}_O} |P_0(B) - P_1(B)|$. Under equal prior weight, every measurable, possibly randomized adjudicator has error at least

$$\frac{1 - \text{TV}(\mathcal{L}(O \mid W_0), \mathcal{L}(O \mid W_1))}{2}. \quad (3)$$

The infimum error over adjudicators equals this bound. For two probability measures it is attained: a Hahn decomposition of the finite signed measure $P_1 - P_0$ supplies a maximizing acceptance region. In particular, if the observable records have the same distribution, the competence claim is not identifiable and the error is exactly $1/2$ for every adjudicator.

Proof. For any acceptance region B , the equal-prior error is $\{P_0(B) + P_1(B^c)\}/2$. Taking the infimum over B gives one half of one minus the total-variation distance. The positive set in a Hahn decomposition of $P_1 - P_0$ attains the supremum, while randomized acceptance probabilities cannot improve on it because their risks are mixtures of deterministic risks. $\square$

This second bound concerns interpretation rather than terminal prediction. A score can separate output interfaces while leaving two explanations of that score observationally equivalent. For example, one rule may recognize an epistemic insufficiency, whereas another may be forced into the only available non-promoting terminal. A broader competence headline remains unidentified until a discriminator changes the observable law of those worlds.

Nuisance identifiability. Let N contain construction-only quantities such as list length, missingness, template, key order, path name, or label-bearing metadata. The nuisance-only Bayes advantage is

$$I_a(N) = \mathbb{E}_N \left[\max_t \Pr(Y_a = t \mid N) \right] - \max_t \Pr(Y_a = t). \quad (4)$$

A scientific interpretation requires the intended semantic representation to make the terminal attainable while nuisance information contributes no material advantage. A finite shortcut register cannot certify $I_a(N) = 0$ for every possible decoder. It certifies only that the frozen probe class found no such advantage, which is why claim-axis authorization must name the probe class and cannot be promoted to whole-register or naturalistic clearance.

Panel resolution and interface attainability. For a frozen panel $\mathcal{H}$ and score M_a , define its observed resolution as

$$\rho_a(\mathcal{H}) = \max_{h \in \mathcal{H}} M_a(h) - \min_{h \in \mathcal{H}} M_a(h).$$

When $\rho_a = 0$, the experiment supplies no comparative ordering on that axis; it does not establish equality of the underlying competences. When $\rho_a > 0$ but some comparators lack a target terminal in their output alphabet, the contrast establishes interface expressiveness before it establishes the finer judgement of when that terminal is scientifically appropriate. A broad verification-axis claim is therefore licensed only inside the intersection of four conditions: semantic attainability, nuisance resistance, matched terminal attainability, and nonzero panel resolution. Failures fall into distinct research programmes rather than one generic null: repair the construction, enlarge the interface, expose a target-sufficient representation, or improve the decision rule.

Application to the three P4 evidence layers. The historical V2 H3 axis had $\rho_a = 0$: an empty evidence list exposed the gold family and every panel member scored 30/30. That result remains NOTSUPPORTED. The later V3 construction removes the registered shortcuts and obtains nonzero resolution, but nine of ten comparator mechanisms cannot emit CANNOTCHECK; its exact claim is therefore terminal expressiveness under the frozen gate lattice, not general epistemic judgement.

Stated as an estimand: where a comparator interface cannot emit CANNOTCHECK, the quantity this comparison estimates is *terminal attainability* on the axis, and nothing else. A system that cannot express the correct terminal cannot be observed to reach it, so a gap measured against such an interface is a fact about what the interface can say, not about what the system knows. Reading it as epistemic competence would attribute to judgement what belongs to expressiveness. The P4-X exact-contract successor tests the donor-factorization boundary more directly. Its ideal typed donor product receives the target-sufficient coordinates and predicate and ties exactly at 400/400, while the registered generic-authorization and compensatory products do not. These layers diagnose different limits and are not pooled.

4 Non-escalating authority transition

Let a proposal contain claim c and evidence references $E = \{e_1, \dots, e_n\}$. Promotion is a conjunction of hard prerequisites rather than a weighted score:

$$G_{\text{content}} : \text{resolved bytes equal the registered evidence digest}, \quad (5)$$

$$G_{\text{owner}} : \text{the assigned source owns the cited content}, \quad (6)$$

$$G_{\text{support}} : \text{the assigned evidence supports } c, \quad (7)$$

$$G_{\text{indep}} : \text{checker lineage is admissibly independent}, \quad (8)$$

$$G_{\text{disc}} : \text{checker rejects the frozen hostile battery}, \quad (9)$$

$$G_{\text{influence}} : \text{cited evidence participated in the answer path}, \quad (10)$$

$$G_{\text{eval}} : \text{evaluator identity/chronology is intact}, \quad (11)$$

$$G_{\text{access}} : \text{protected-label access telemetry is clear}, \quad (12)$$

$$G_{\text{search}} : \text{search-time contamination is clear}. \quad (13)$$

The claim is promoted only when every required gate passes and the evidence supports it. A resolved negative prerequisite blocks promotion; an unresolved evidential prerequisite leaves the claim undetermined. No failed coordinate can be averaged away by confidence elsewhere. Eight ablations remove one coordinate at a time, including a soft-confidence variant that promotes when at least six of nine gates pass.

5 Methods

5.1 Three-valued outcomes

Every authority decision this paper measures is three-valued rather than binary. A claim may be promoted, blocked, or left *undetermined*. An outcome that could not be determined is recorded and counted separately from an outcome determined to be false, and the two are never merged. The distinction is what makes a missing prerequisite legible: an unresolvable digest, an unreachable checker, or an absent provenance record leaves the claim undetermined, and an undetermined claim is not a refuted one. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count.

5.2 The implementation under test

The implementation evaluated here is the ORION authority layer; it is the artifact under test, and the rest of the paper calls it the governed pipeline so that each claim reads as a claim about the mechanism rather than about a product. The results tables and figures label its rows and points by that implementation name because they are projections of the archived campaign record. Protocol and evidence materials are described in Section 10.

5.3 Prospective freeze and fresh post-repair split

The base protocol, `P4.protected-authority.v1`, was frozen prospectively, and the post-repair campaign reported here is an additive execution binding on top of it, with a fresh hidden split and a repaired panel. Before any outcome-bearing run, a public-seed preflight exercised 42 hostile authority tests and the complete 420-case pipeline with ten comparator mechanisms, eight ablations, five deterministic repeats, protected scoring, typed-panel validation, and a separately implemented reproducer. After the preflight, execution binding, and harness had been frozen, a separate isolated run minted a fresh secret-seeded split. The secret seed was never exposed to the candidate or comparator lanes; only its commitment was retained.

The final split contains 420 mechanical-gold cases across 13 families: 60 clean positives and 30 cases in each of 12 hostile or insufficient-evidence families. A 150-case post-run public slice contains 30 clean and 120 hostile cases; distinct protected-hostile and protected-holdout slices remain hidden during scoring. Table 1 summarizes custody.

Table 1: Protected V2 battery and custody. Hidden labels remained host-only until scored jobs completed.

Slice	Cases	Clean	Hostile	Candidate labels
Post-run public slice	150	30	120	hidden during execution
Protected hostile	120	0	120	hidden
Protected holdout	150	30	120	hidden
Total	420	60	360	hidden

5.4 Gold-blind execution and custody

The candidate and comparator lanes received exactly two top-level fields per case: an opaque case identifier and the candidate-visible packet. Before execution, the host mechanically checked that attack family, protected gold, custody class, and expected authority terminal were absent. Candidate and comparator lanes then ran in isolated environments without project-source access credentials. Five deterministic repeats were retained under file and network syscall tracing. The protected evaluator became eligible to access gold only after both scored lanes completed, and first checked that the retained traces contained neither protected identifiers nor external-IP connections.

Immediately before handoff, the launch controller revalidated the signed execution-freeze revision and the corresponding continuous-integration attestations, then byte-compared the frozen execution, binding, and harness inputs against that revision. The evaluator and the separately implemented reproducer ran as distinct code paths. These controls establish process and implementation separation inside one project; they do not constitute institutionally independent evaluator or result-verifier custody.

5.5 Comparator panel, ablations, and statistics

Seven primary comparator mechanisms mirror the repaired panel: ProvenanceGuard-style source routing, AttributionBench-style multi-source attribution, FIRE-style iterative retrieve/verify, CLAIM-BENCH-style claim evidence, ProvenAI-style citation fidelity/influence, RewardHackingAgents-style contamination/tamper checks, and DeepSciVerify-style evidence escalation. Three auxiliary controls test citation presence, pooled semantic support, and auditability/provenance.

Comparator names therefore use “-style” or describe a mechanism. They are protocol-matched reimplementations under one candidate-visible packet and resource accounting, not executions of the external authors’ software. Every empirical comparison below is about the frozen mechanisms in this benchmark, not about published systems in their native environments.

The primary outcome is false authority-promotion rate over gold terminals that do not permit promotion. Clean coverage is promotion rate over the 60 clean positives. H1 requires an absolute false-promotion improvement beyond 0.05 against the strongest frozen comparator; H2 requires clean coverage non-inferior within 0.05; H3 tests superior selection of the undetermined terminal where that is the correct answer. Rate intervals use Wilson 95% intervals; paired effects use a fixed-seed percentile bootstrap. No false promotion, false block, abstention, candidate-caused failure, or clean false negative is removed from its eligible denominator.

6 Results

The typed authority panel passed with no blockers, and a separately implemented local reproducer obtained the same headline counts from the safe publication bundle. This is code-path separation, not external verification authority.

6.1 False promotion and clean coverage

The governed pipeline produced 0/360 false promotions. The strongest frozen comparator mechanism, ProvenAI-style citation-fidelity/influence, produced 180/360 (50.0%). The paired difference was -0.50 with

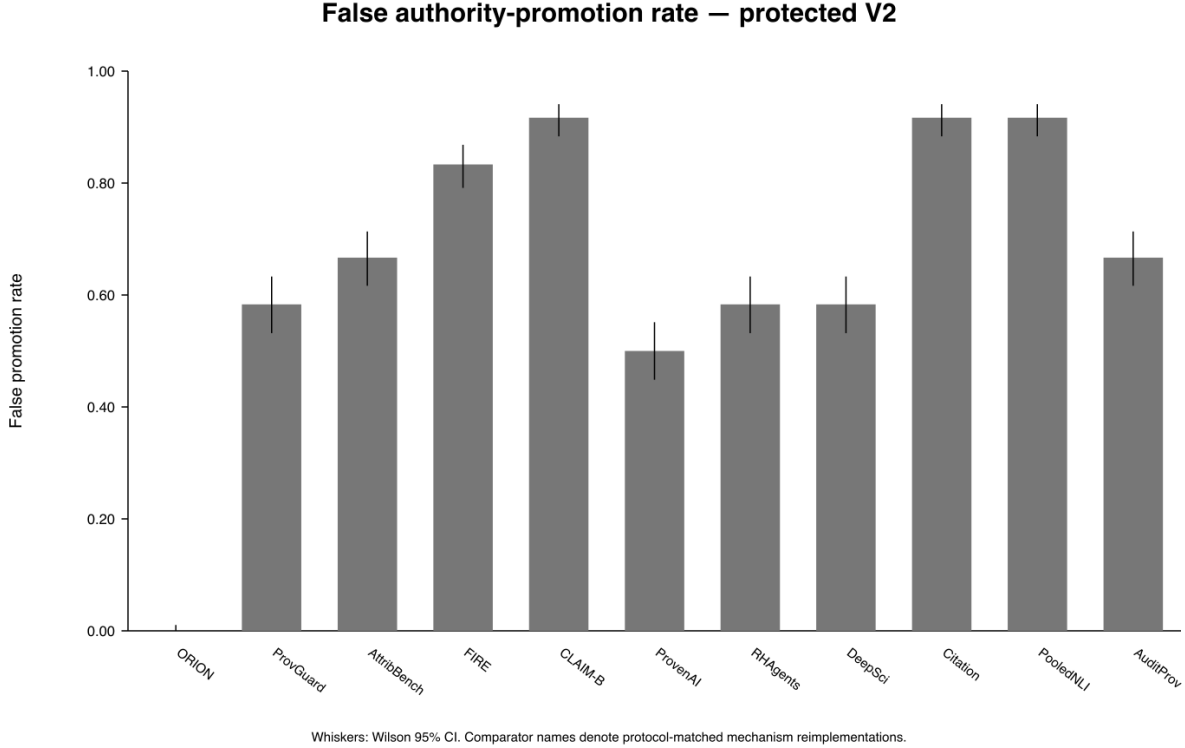


Figure 1: False authority-promotion rate over 360 hostile opportunities in the protected V2 campaign. Whiskers are Wilson 95% intervals for each rate. Comparator labels denote protocol-matched mechanism reimplementations, not the original external systems.

95% CI $[-0.553, -0.447]$, so H1 passed the predeclared -0.05 practical margin. Figure 1 reports the full mechanism panel.

Both the governed pipeline and the strongest comparator promoted 60/60 clean positives, with zero clean false negatives. The clean-coverage difference and paired 95% interval were both exactly zero, so H2 passed the -0.05 non-inferiority margin. Figure 2 shows that every primary mechanism sits at 1.0 clean coverage; the result therefore does not manufacture a frontier by visually jittering equal values.

On this battery H3 was not supported, and the cause is in the battery rather than in the systems. Every one of the eleven panel members, the governed pipeline included, correctly left all 30 insufficient-evidence cases undetermined, so the difference is zero with 95% CI $[0, 0]$, and the ablations do not instrument the metric at all.

The construction explains why. In this battery the insufficient-evidence family was generated with an *empty* evidence list, while every other family carried exactly one evidence object. The rule “leave the claim undetermined if and only if the evidence list is empty” therefore classifies all 420 cases correctly. Detecting an empty list is not an act of judgement: this construction saturates the metric, leaving the comparison with no resolving power, so a difference could not have been observed had one existed. We report this as an inability to discriminate rather than as evidence that no difference exists, and we do not treat the null as a finding about abstention. Table 2 reports abstention and false-negative cost on that basis.

6.2 Repairing the construction, and one run of the frozen panel on it

The first repair changed what must be reasoned about rather than any gold label: cases present evidence that is read but does not establish the claim, in two subtypes, one where the pool contains no supporting record

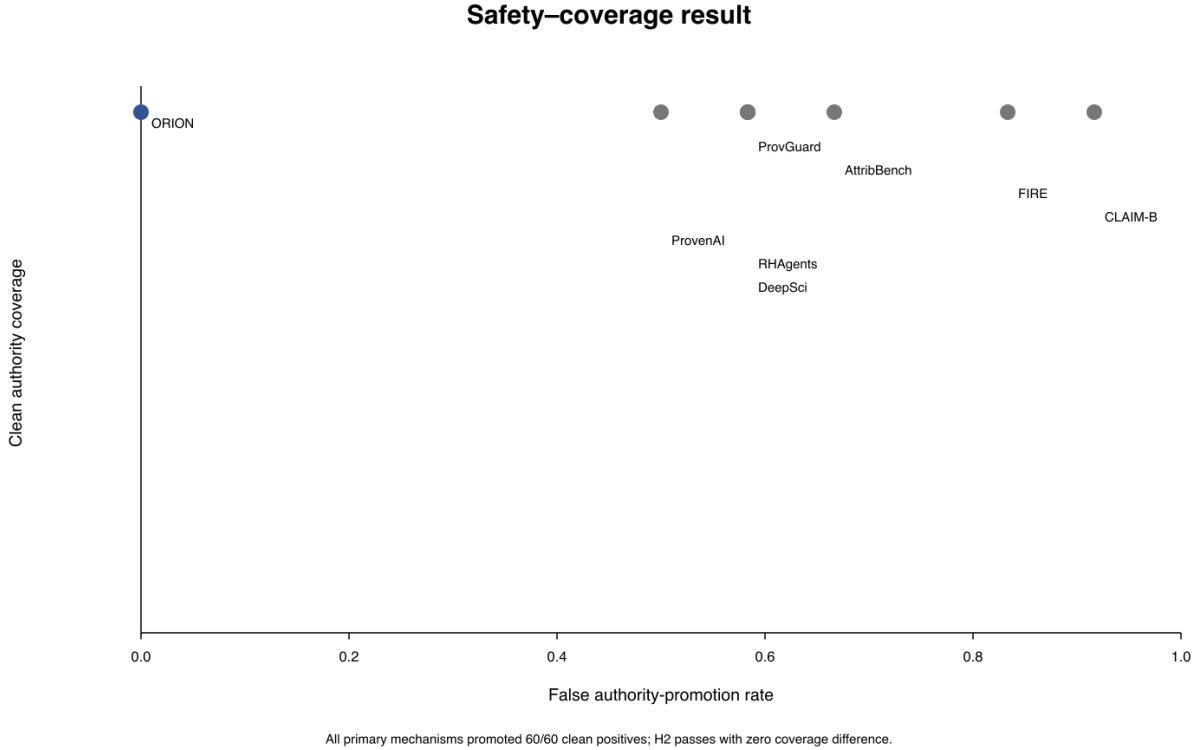


Figure 2: Safety–coverage result in protected V2. Each point combines false-promotion rate over 360 hostile opportunities (horizontal axis) with promotion coverage over the same 60 clean controls (vertical axis). Every primary mechanism promoted 60/60 clean controls; H2 therefore has zero paired coverage difference and a degenerate 95% interval at zero, while hostile false-promotion rates differ.

Table 2: Fail-closed outcomes and false-negative cost in protected V2. The only gold family whose correct terminal is ‘undetermined’ contains 30 insufficient-evidence cases.

System	Correctly undetermined	False promotions	Clean FN
ORION	30/30	0/360	0/60
ProvenAI-style	30/30	180/360	0/60

at all and one where a supporting record is present but carries no verifiable provenance. That removed the empty list and did not remove the shortcut. The repaired bodies took five distinct lengths, two of which occur only on undetermined cases, so a rule over evidence-body length recovers the terminal on 20/20 protected cases with no false positives among the other 250 — and the family named itself in English besides. Measured after the fact with the register described below, five of fourteen judgement-free probes recover the undetermined label at informedness 1.00 on that construction, and thirteen of the fourteen do so on the published one. Repairing a named shortcut is not the same as establishing that there is none, so we do not report a panel from it.

The second repair was written against the property rather than against a named cue, and its protocol, seeds and interpretation were frozen in advance of the repair and of any panel outcome. Every case in the battery is shape-identical: one evidence-body template for all 840 records, with the same length, word count, punctuation and character-class profile; fixed container sizes throughout; no absent field anywhere; and support carried by a token that is present on every record, so that only token matching separates support from non-support. The undetermined family is split into two shape-identical subtypes and is a strict subset of a set that also contains 30 blocking cases, so neither “nothing in the pool supports” nor “the assigned

record does not support” classifies it. Families differ only in the values of the fields an authority gate is defined over. The identifiability register — fourteen judgement-free probes over counts, container lengths, key shapes, string lengths, character-class and template fingerprints, identifier shapes and scalar values, fitted on the fitting split and scored on the protected one — was run and recorded before the panel was invoked. On the repaired construction it reports informedness 0.0 for every probe on the undetermined axis, on all thirteen registered seeds, against a declared ceiling of 0.0; not one probe ever predicts the undetermined terminal for any protected case.

Run once on that battery, the same frozen panel and statistics report H3 supported. The governed pipeline selects the undetermined terminal on 30/30 eligible cases with 0/360 false promotions; the strongest frozen comparator, selected by H1 exactly as before, selects it on 0/30; the paired difference is 1.0 with 95% CI [1.0, 1.0] against a predeclared threshold of a positive lower bound. The one comparator mechanism with an escalation path, DeepSciVerify-style abstract-to-full escalation, selects it on 15/30, so the margin against that mechanism is 0.5 rather than 1.0; both numbers are needed to read the result. What the difference measures was fixed before the panel ran and is narrower than the headline: terminal expressiveness under a non-compensatory gate lattice — the ability to report an inability — and not a finer-grained scientific judgement. Nine of the ten comparator mechanisms score zero because, once the empty-evidence case is gone, they cannot express an undetermined terminal at all, and a panel of eleven systems in which ten are two-valued cannot separate that reading from a claim about knowing when one cannot check. Clean coverage is 1.0 for all eleven systems on the repaired battery as well, so H2 remains a saturated axis there and is still not a comparative finding. The V2 numbers above are unchanged and are not restated by this run: it is a different battery, and its result does not license reinterpreting the null the V2 construction produced.

6.3 Attack families and ablations

The governed pipeline made no false promotion in any hostile family. The strongest comparator false-promoted all 30 cases in each of six governance/checker families: weak checker, same-lane checker, post-hoc checker/evaluator, search-time contamination, evaluator tamper, and holdout access. It was correct on content substitution, source conflation, wrong-source, pooled-owner, cited-non-influential, and insufficient-evidence families. Figure 3 shows the contrast.

Every registered ablation preserved 60/60 clean coverage while increasing false promotion. Removing exact content binding, checker-lineage independence, hostile-checker discrimination, behavioral influence, or the search-contamination block produced 30/360 false promotions (8.3%) each. Collapsing source/provenance identity produced 60/360 (16.7%); removing evaluator protection/telemetry produced 90/360 (25.0%); and replacing the fail-closed terminal with the predeclared six-of-nine soft-confidence threshold produced 330/360 (91.7%). Table 3 contains the complete panel.

6.4 Attribution, support, cost, and telemetry

Several mechanisms achieve perfect source-attribution accuracy while still false-promoting governance attacks; conversely, pooled semantic-support mechanisms can score well on support while attributing the source incorrectly. Figure 4 separates those coordinates.

Deterministic gate evaluation is slower than the simple mechanism proxies in this synthetic harness: mean latency is about 2.97 ms per case for the governed pipeline versus single-digit microseconds for the primary proxies (Figure 5). This is not an end-to-end serving benchmark; model and network latency are intentionally absent.

Scored-process telemetry recorded zero protected-identifier hits and zero external-IP connections for both candidate and comparator lanes. Raw traces remain in protected custody. The separate local reproducer obtained the same 0/360 versus 180/360 false-promotion counts and 60/60 versus 60/60 clean-promotion counts; this code-path separation does not supply external result-verifier custody.

	ORION	ProvenAI
Wrong Source	0%	0%
Content Substitution	0%	0%
Source Conflation	0%	0%
Pooled Support Wrong Owner	0%	0%
Cited Non Influential	0%	0%
Weak Checker	0%	100%
Same Lane Checker	0%	100%
Posthoc Checker Evaluator	0%	100%
Search Time Contamination	0%	100%
Evaluator Tamper	0%	100%
Holdout Access	0%	100%
Insufficient Evidence	0%	0%

ProvenAI-style false-promotes all 30 cases in six governance/checker families; ORION false-promotes none.

Figure 3: False-promotion rate by hostile family in protected V2 for the governed pipeline and the strongest frozen mechanism proxy. Each row contains 30 cases; the governed pipeline false-promoted none, whereas the proxy false-promoted all cases in six governance/checker families.

6.5 Exclusion of exploratory evidence

We exclude an earlier 39-case live-model arm from publication authorization. Post-run audit found internally inconsistent live labels/expected terminals, incorrect opportunity denominators in its summary path, and direct use of a candidate-hidden family label in the live execution path for the governed pipeline and its ablations. Those outputs remain useful diagnostics but are not combined with, substituted for, or used to tune the protected campaign.

Table 3: Protected V2 results. Comparator names denote protocol-matched mechanism reimplementations, not original external software. FP is over 360 hostile opportunities; clean is over 60 clean positives.

System / variant	FP	Clean	Clean FN	Latency (ms)	Resource
ORION	0.000	1.000	0	2.9736	9.0
ProvenanceGuard-style	0.583	1.000	0	0.0037	5.0
AttributionBench-style	0.667	1.000	0	0.0058	3.0
FIRE-style	0.833	1.000	0	0.0069	6.0
CLAIM-BENCH-style	0.917	1.000	0	0.0049	4.0
ProvenAI-style	0.500	1.000	0	0.0034	5.0
RewardHackingAgents-style	0.583	1.000	0	0.0072	5.0
DeepSciVerify-style	0.583	1.000	0	0.0074	6.5
Citation presence	0.917	1.000	0	0.0009	1.0
Pooled support	0.917	1.000	0	0.0049	2.0
Auditability/provenance	0.667	1.000	0	0.0033	5.0
ORION - exact content	0.083	1.000	0	2.7131	9.0
ORION - source/provenance	0.167	1.000	0	2.7077	9.0
ORION - checker lineage	0.083	1.000	0	2.7078	9.0
ORION - hostile checker	0.083	1.000	0	2.7043	9.0
ORION - influence	0.083	1.000	0	2.7045	9.0
ORION - evaluator protection	0.250	1.000	0	2.7116	9.0
ORION soft confidence	0.917	1.000	0	2.4570	9.0
ORION - search block	0.083	1.000	0	2.7087	9.0

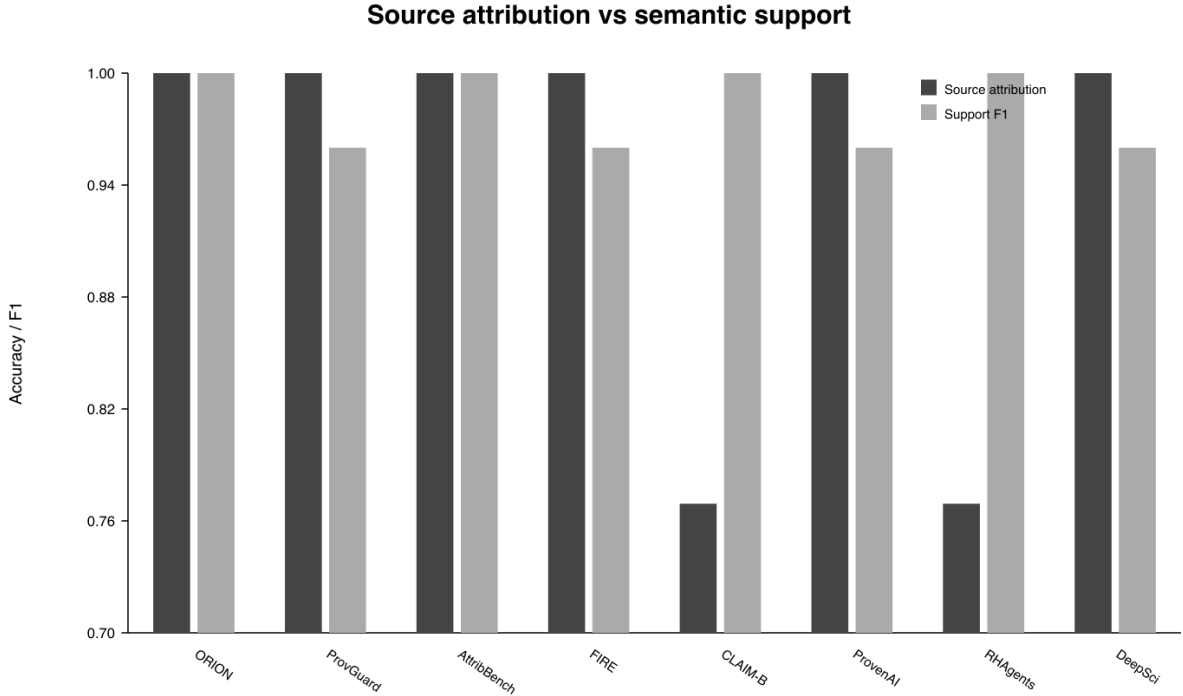


Figure 4: Source-attribution accuracy and support/contradiction F1 for the primary mechanism panel on the protected 420-case V2 battery. Source attribution tests assignment to the correct evidence owner; support/contradiction F1 tests the semantic evidence label. High values on either coordinate do not guarantee a safe authority terminal.

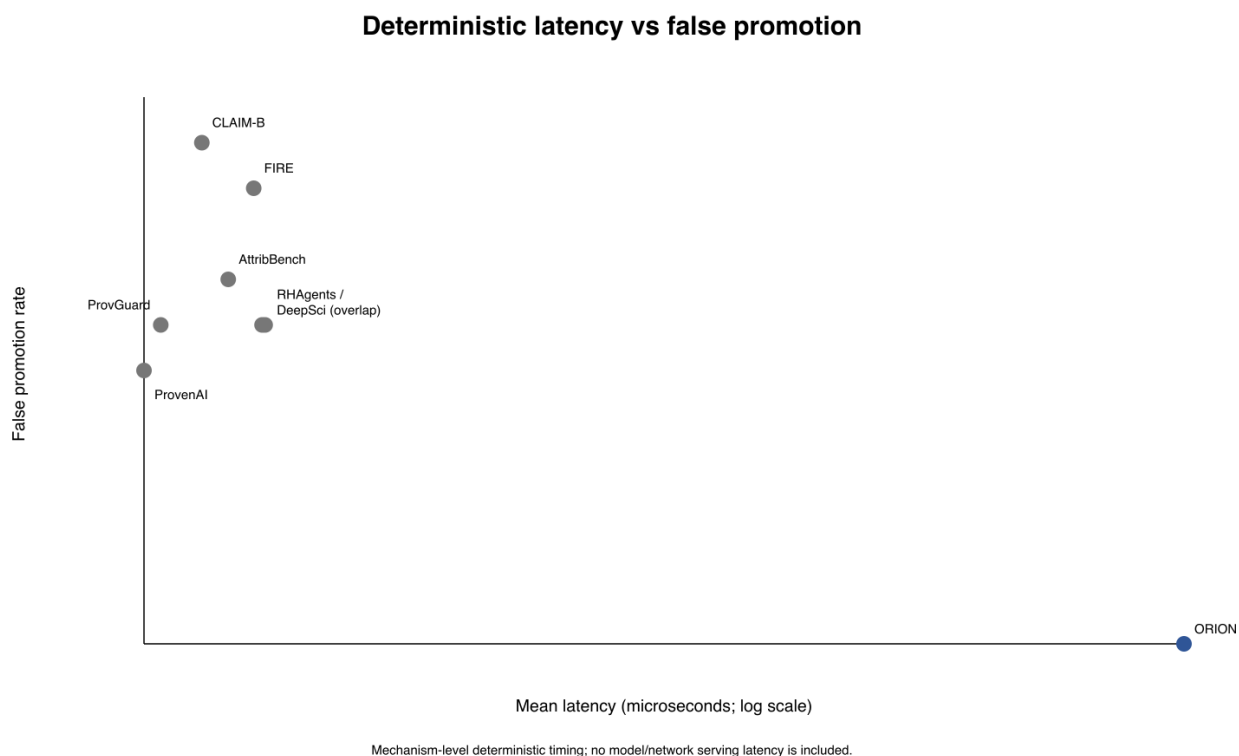


Figure 5: Mean deterministic evaluation latency across five repeat runs versus false authority-promotion rate over 360 hostile opportunities in protected V2; the latency axis is logarithmic. Timing covers mechanism-level gate evaluation only and excludes model inference, network transport, and end-to-end serving overhead.

7 Post-saturation successor: donor-complete scientific-promotion authority

The protected V2 result shows that a non-compensatory content/provenance/checker/evaluator pipeline can eliminate false scientific-authority promotion relative to strong frozen mechanism proxies on the registered battery while preserving every clean positive. P4-X raises the comparison standard. Ordinary provenance, claim verification, artifact/version binding, evaluator custody, epoch handling, and generic authorization are granted to a donor-complete product, leaving only the final *scientific-promotion relation* as the discriminator. The V2 and successor campaigns remain separate prospectively frozen experiments and are not pooled.

7.1 Donor-engulfment architecture

P4 retains source-aware factuality, provenance, claim/evidence verification, citation fidelity, evaluator auditing, contamination defense, abstention, and generic authorization as donor-owned local components (Alvarez et al., 2026; Faizan & Alharthi, 2026; Liao, 2026; Tu et al., 2026; Zhou et al., 2026). P4-X tests the higher-level authority relation that determines whether those successful local verdicts are sufficient to change the scientific standing of a target claim.

The P4-X relation is non-compensatory. A target claim is promoted only when every registered hard scientific-promotion obligation is discharged. In the exact successor battery, the discriminating coordinates concern target claim scope, independence of supporting evidence, and unresolved scientific-promotion authority after the ordinary provenance, verification, custody, and authorization checks available to the donor product have already succeeded.

7.2 Exact successor design

The frozen successor contains 400 protected cases spanning five heterogeneous scientific artifact domains, eight hostile/control archetypes, and ten protected variants per domain–archetype cell. All systems receive the same candidate-visible information.

The principal arms are:

- **P4-X**: the non-compensatory scientific-promotion relation;
- **B1**: a strong donor-complete generic-authorization product with ordinary provenance, verification, artifact/version, custody, and epoch information but without the complete target-bound scientific-promotion relation;
- **B2**: a compensatory all-signal product that permits successful coordinates to offset failed or unresolved ones;
- **B3**: an ideal typed donor product given every P4-X scientific coordinate and the same promotion predicate, testing portability of the semantics.

7.3 Protected result: perfect exact promotion decisions

P4-X selects the correct scientific-promotion terminal on **400/400 protected cases**. B1 is correct on 250/400, B2 on 50/400, and B3 on 400/400 with zero decision mismatches to P4-X. The paired P4-X minus B1 effect is +0.375 with a domain-stratified bootstrap 95% CI [0.3275, 0.4225]; paired McNemar discordance is 150–0. P4-X makes **zero false promotions**, retains clean promotion at 1.0, and its advantage over B1 is positive in every exact domain. An independent implementation that does not import the P4-X execution scorer reproduces the canonical protected-row digest and all arm counts.

The result isolates a specific decision difference under the registered exact contracts. Ordinary provenance, verification, custody, and generic permission can all be valid while claim scope, evidence dependence, or an unresolved scientific obligation still prevents promotion. In this battery, making the target-bound promotion relation explicit closes all 150 B1 errors without sacrificing any clean promotion.

7.4 Portability result

B3 also reaches 400/400 and is decision-identical to P4-X. This is a positive systems result: the scientific-promotion semantics are sufficient and portable within the registered exact-contract representation. They can be implemented as an ORION layer or as a correctly enriched donor product without changing any registered decision. The P4-X advantage over B1 therefore identifies the missing scientific coupling in the registered donor-complete interface rather than depending on centralized expressivity.

7.5 Measurement-resolution audit

The successor does not retroactively reinterpret the historical V2 secondary metrics. V2 H2 is a clean-coverage guard in a battery where all primary systems achieve perfect clean coverage, confirming that the false-promotion result is not blanket refusal. V2 H3 is non-discriminating because the original insufficient-evidence construction is structurally saturated. The post-run audit identifies that shortcut and prevents a trivial equality from becoming a scientific equivalence claim. This instrument diagnosis strengthens the authority discipline of the paper while leaving the protected V2 and P4-X headline results unchanged.

7.6 Strongest supported statement

After donor-complete provenance, verification, custody, artifact binding, freshness, and generic authorization are granted, *scientific claim promotion remains a distinct non-compensatory authority relation in these registered exact contracts*. Making that relation explicit raises exact promotion accuracy from 0.625 to 1.000 against the strong donor-complete product while preserving zero false promotions and perfect clean coverage. Live verifier/provider transfer is the next empirical stress test of this demonstrated bounded relation.

8 Threat model, limitations, and interpretation

The theory and the evidence have different scopes. Proposition 1 and its factorization corollary hold for finite exact-terminal decision problems, but they identify when a target is attainable from a representation and output alphabet; they do not show that a deployed verifier possesses a target-sufficient representation. The empirical layers establish only the frozen representations, relations, and case distributions evaluated here.

First, the V2 and V3 campaigns use 420 synthetic mechanical-gold cases rather than naturally occurring scientific disagreements. The written human-adjudication rubric was never triggered because all final cases were mechanically decidable, so no claim is made about expert agreement. P4-X broadens the exact-contract battery to five artifact domains and eight hostile/control archetypes, but its 400 cases are still constructed contracts, not a population sample of scientific disputes. The final scoring logic is deterministic and uses no language-model provider; it isolates authority-transition semantics rather than the error distribution of a full stochastic research agent.

Second, the comparator mechanisms are matched reimplementations. They instantiate published ideas under one candidate-visible packet and common accounting, but the results are not a leaderboard against the original ProvenAI, ProvenanceGuard, DeepSciVerify, FIRE, AttributionBench, CLAIM-BENCH, Reward-HackingAgents, or related systems in their native environments. Likewise, 60/60 clean coverage shows that the governed pipeline did not obtain its V2 false-promotion advantage by blanket refusal; the clean controls are intentionally unambiguous and do not estimate recall on difficult benign cases. The hostile taxonomy is finite, so adaptive attacks, compromised host infrastructure, unobserved side channels, and novel evaluator attacks remain outside the campaign.

Third, two registered V2 axes are saturated and must be read as design limits rather than comparative findings. Clean coverage is 1.0 for all eleven primary systems and every registered ablation. The original H3 family is also saturated: an empty evidence list makes every panel member correct on all 30 eligible cases. H1 therefore supplies the V2 comparative measurement, whereas H2’s non-inferiority pass and V2 H3’s null do not establish equality of underlying competences. The V3 construction repairs the registered H3 shortcuts and produces nonzero panel resolution, but nine of ten comparator mechanisms cannot emit the

undetermined terminal. Its result therefore measures terminal/interface attainability under the frozen gate lattice. The 1.0 margin against the H1-selected comparator and the 0.5 margin against the escalation-capable mechanism are both retained precisely to prevent a broader judgement interpretation.

The nuisance audit has the same bounded scope. Fourteen frozen probes, evaluated across thirteen registered seeds, found no advantage on the exact V3 undetermined axis. That is evidence against the declared shortcut class, not a proof that no possible nuisance decoder exists. More generally, the theory separates semantic attainability, nuisance resistance, matched terminal attainability, and panel resolution because failure of any one of them changes what a benchmark contrast can mean.

Fourth, the component mechanisms are not claimed as novel. Provenance tracking, attribution evaluation, citation checking, iterative verification, evidence escalation, contamination detection, evaluator-tamper detection, abstention, benchmark auditing, claim-level auditability, provenance ontologies, and generic behavioral-integrity checking all have prior art discussed in Section 2. The contribution tested here is the non-compensatory composition of these prerequisites and, in P4-X, the target-bound scientific-promotion relation above a donor-complete generic authorization product. The information-equivalent typed product's 400/400 tie is a positive boundary result: the evidence does not support a centralization advantage, universal necessity of the exact gate list, or superiority over deployed donor systems.

Finally, naturalistic transfer remains unestablished. A prospective programme specifies 768 source-case-family clusters across four domains and eight information-state mechanisms, paired identifiable controls, worst-domain analysis, source-disjoint replication, and independent evaluator custody. The programme has so far produced only source-feasibility and identity-transport evidence. Its strongest current public identity bridge reaches 76/80 within one bounded software-publication frame, while author-lineage adjudication, same-claim natural-pair eligibility, source-disjoint replication, external custody, and exact external-comparator execution remain unresolved; zero natural pairs have been authorized. These preflights localize what a future naturalistic study would need, but they do not contribute performance evidence to the present paper. The detailed transport history is retained in the prospective supplementary record rather than used as part of the main decision proof.

Computational integrity is also not scientific authority. Frozen digests, deterministic replays, custody controls, and independent code paths establish that a stated computation was the computation performed and can be reconstructed under the released conditions. They do not by themselves establish that the benchmark measures the intended construct or that an external party has independently validated the scientific interpretation. A perfectly replayed flawed protocol would reproduce the flaw exactly; this is why the V2 H3 construction failure remains visible rather than being repaired retrospectively.

9 Broader impact and governance

False scientific-authority promotion can amplify incorrect or misattributed claims, particularly when a system presents verification artifacts that appear independent but are not. A fail-closed layer may reduce this risk, but excessive blocking can delay legitimate scientific communication or concentrate power in whoever controls evidence and evaluator registries. Work on academic integrity and agentic abstention reinforces that refusal can be the correct behavior under genuine insufficiency, while also showing that productive abstention remains difficult and should not be conflated with generic non-response (Yang et al., 2026; Liu et al., 2026). We therefore report abstention cost, clean coverage, custody identity, and artifact transparency rather than optimizing only a security metric.

Protected evaluation creates another trade-off: some labels must remain hidden during scoring, yet publication requires enough evidence for audit. The evidence package retains safe aggregates, a post-run public slice, exact subject/split/harness bindings, and a separate-implementation reproduction receipt while protected holdout labels and raw traces remain in restricted custody. This receipt supplies local code-path separation, not external result-verifier custody.

10 Data and code availability

The anonymous review package contains safe aggregate result objects for the protected V2 battery, the distinct V3 exact-axis battery, and the P4-X exact contracts; canonical-source checksums; a member checksum list; and a standalone verifier for the headline counts and retained interpretation boundaries. The package preserves the V2 H3 status as not supported and keeps the three empirical layers separate rather than pooling them.

Protected per-case gold, raw execution traces, secret seeds, and candidate-hidden fields are withheld from the review bundle because they are part of the custody model under test. Their absence is explicit rather than treated as public reproduction. The packaged verifier therefore checks the published safe aggregates and headline counts; it neither reruns protected scoring nor reconstructs hidden gold from information unavailable to the candidate or comparator lanes.

For double-blind review, identifying public-repository metadata and author-linked archive information are kept out of the manuscript-facing access path. The exact review-artifact members are checksum-bound and can be uploaded with the paper as anonymous supplementary material. Public repository and archival identifiers can be restored in the non-blinded release package without changing the scientific result. No persistent DOI is asserted in this review manuscript.

The naturalistic source-expansion programme is not part of the evidence supporting the present performance claims. Its outcome-free source and transport records are retained separately as prospective supplementary material so that an unexecuted successor is distinguishable from an omitted experiment.

11 Conclusion

A scientific-verification result requires more than a score. The intended terminal must be identifiable from admissible scientific information, resistant to the nuisance class being tested, expressible by the compared interfaces, and capable of varying over the panel. The exact-risk and factorization results make these requirements explicit: a target relation is implementable when it factors through the decision representation and its required terminals are available, whereas mixed fibres impose irreducible error.

The three empirical layers instantiate different parts of that theory without rewriting one another. V2 observed 0/360 false promotions for the governed pipeline versus 180/360 for the strongest frozen mechanism comparator while both promoted 60/60 clean positives; its original H3 family was construction-saturated and remains NOTSUPPORTED. V3 removed the registered shortcut class and separated the undetermined terminal, but because most comparators cannot emit that terminal, the supported result is one of interface/terminal attainability rather than general scientific judgement. P4-X then raised the comparison standard: after donor-complete provenance, verification, custody, version, epoch, and generic-authorization machinery were granted, the target-bound non-compensatory relation scored 400/400 exact decisions versus 250/400 for the registered generic product and 50/400 for a compensatory product. An information-equivalent typed product also scored 400/400, showing that the semantics are portable rather than inherently centralized.

The contribution is therefore wider than a gate checklist but narrower than a deployed-system victory. It provides a theory for deciding when verification axes can support scientific claims and finite protected evidence showing that local verification and generic permission need not suffice for a target scientific-promotion relation. Naturalistic, source-disjoint, externally custodied evaluation remains open: the prospective source programme has not authorized a single natural pair or executed the external comparator panel. Those successor preflights are retained as prospective records, not promoted into evidence for the present paper.

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